Quantification: 682_z29
=======================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
----

measure	value
lacuna count	13
interior lacuna count	11
frame-edge lacuna count	2

Interior lacunae only: mean, median and sample SD (n = 11)
----

measure	kind	mean	median	sd	units
area_px2	lacuna shape	1740.55	1624	939.642	px^2
major_axis_length_px	lacuna shape	75.1181	75.9278	23.0542	px
minor_axis_length_px	lacuna shape	29.0371	30.0551	9.2359	px
aspect_ratio	lacuna shape	2.7064	2.5263	0.7296	unitless
eccentricity	lacuna shape	0.9096	0.9183	0.0622	unitless
solidity	lacuna shape	0.87	0.8778	0.0425	unitless
roots_count	headline	5.3636	5	1.804	count
ring_length_r30_px	headline	250.454	263	91.0136	px
ring_length_r60_px	reported	647.091	696	247.374	px
owned_length_px	ownership-dependent	1557.3	1262	1443.68	px
edge_count	ownership-dependent	65.9091	48	67.5862	count
mean_edge_length_px	ownership-dependent	25.2613	23.9931	5.9707	px
perimeter_px	normalised	201.309	214.167	60.6733	px
ring_area_r30_px2	normalised	7938.55	8595	2078.18	px^2
ring_area_r60_px2	normalised	19815	20090	4945.82	px^2
in_frame_fraction_r30	normalised	0.9943	1	0.0188	unitless
in_frame_fraction_r60	normalised	0.981	1	0.0583	unitless
ring_density_r30	normalised	0.0313798	0.031331	0.00760002	px^-1
ring_density_r60	normalised	0.0323437	0.0327452	0.00701619	px^-1
roots_per_100px_perimeter	normalised	2.7581	2.7484	0.9127	per 100 px
ring_attached_length_r30_px	network v2	186.909	160	84.6953	px
ring_attached_length_r60_px	network v2	401.636	334	227.182	px
ring_length_w_r30_px	network v2	277.945	289.551	98.9509	px
ring_length_w_r60_px	network v2	723.213	790.642	273.257	px
sholl_crossings_r10	network v2	6.2727	6	2.149	count
sholl_crossings_r20	network v2	9.2727	11	3.6631	count
sholl_crossings_r30	network v2	10.2727	10	3.7707	count
ring_width_mean_r30_px	width	5.9497	6.2323	0.6796	px

Per field
----

measure	value	units
analysed_area_px2	1.02357e+06	px^2
total_skeleton_length_px	29332	px
canalicular_length_density_per_px	0.0286567	px^-1
junction_count	537	count
junction_density_per_px2	0.00052464	px^-2
lacuna_count	13	count
lacunae_per_px2	1.27e-05	px^-2
skeleton_component_count	289	count

mean_component_length_px	101.495	px
median_component_length_px	22	px
field_length_density_w_per_px	0.0324002	px^-1
field_density_without_flagged_per_px	0.0297071	px^-1
field_density_in_roi_per_px		px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	6.7771	px
width_median_px	7.2111	px
width_p10_px	4	px
width_p90_px	10	px
width_pixels_used	28159	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2083	86.508	34.896	2.479	0.915	0.8266	1.5435	13.4983	99.4023	True
2	2213	89.1614	32.1486	2.7734	0.9327	0.8778	-1.1291	134.914	921.292	False
3	446	49.6136	12.2658	4.0449	0.969	0.7866	-1.4445	249.987	449.359	False
4	3140	120.617	34.4194	3.5043	0.9584	0.8885	1.5609	431.048	220.093	False
5	2837	88.9183	41.496	2.1428	0.8844	0.9229	-1.5453	532.043	493.055	False
6	1634	75.9278	30.0551	2.5263	0.9183	0.8211	-1.4797	555.483	357.187	False
7	3781	121.501	43.6088	2.7862	0.9334	0.8771	1.4272	590.482	55.0987	True
8	1399	81.52	23.1918	3.515	0.9587	0.8541	1.5542	716.135	181.57	False
9	513	41.8409	17.3844	2.4068	0.9096	0.8607	1.3405	830.565	75.5867	False
10	1130	64.3164	23.6688	2.7173	0.9298	0.8835	1.4062	842.292	125.237	False
11	1244	49.7173	33.3792	1.4895	0.7411	0.9325	-1.542	904.6	760.807	False
12	2966	92.1224	41.6334	2.2127	0.8921	0.895	-1.4899	920.401	307.385	False
13	1624	72.5441	29.7651	2.4372	0.9119	0.8476	-0.945	978.497	437.615	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	1	128	230	20.4853	1	20.4853	
2	8	310	696	2661.22	110	24.1929	6.5551
3	4	263	566	2166.42	105	20.6325	6.3915
4	7	364	1180	5067.8	236	21.4737	6.3155
5	4	261	753	1884.29	66	28.5499	5.5492
6	6	380	824	1943.44	81	23.9931	6.2323
7	8	348	753	1240.59	48	25.8456	6.0784
8	5	263	717	1262	48	26.2918	4.681
9	5	133	421	419.635	18	23.313	5.633
10	4	155	438	476.889	12	39.7408	7.0548
11	2	118	384	164.581	9	18.2867	5.2457
12	7	320	788	789.612	26	30.3697	5.5407
13	7	188	351	294.421	14	21.0301	6.2476

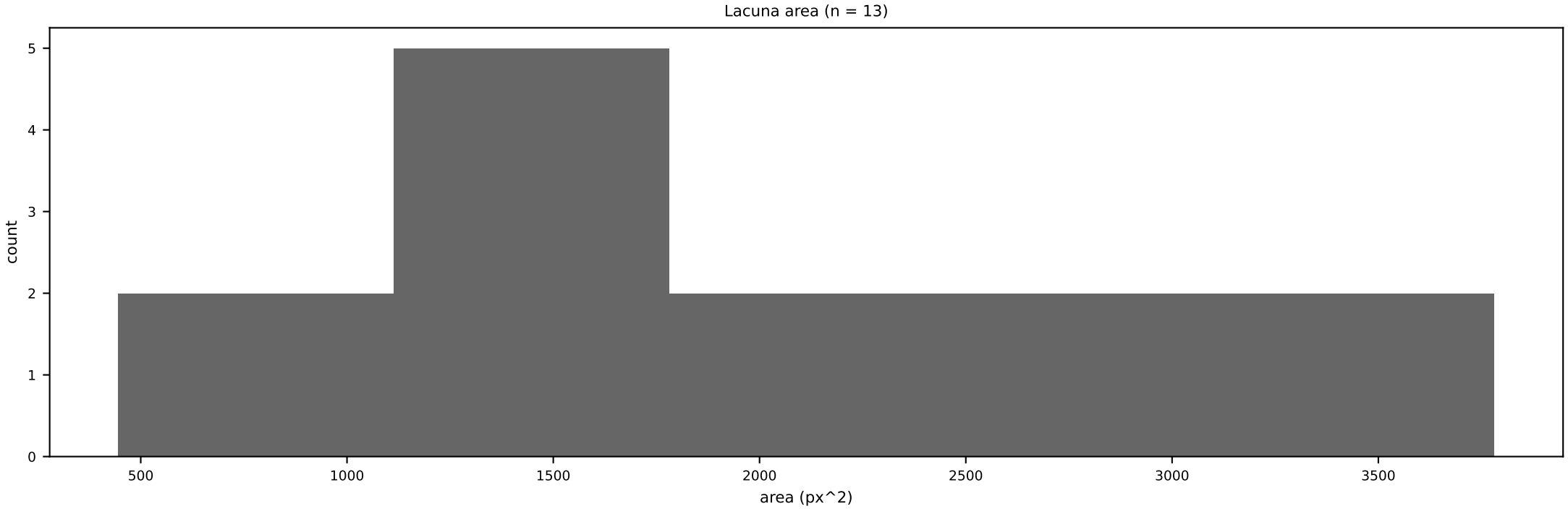
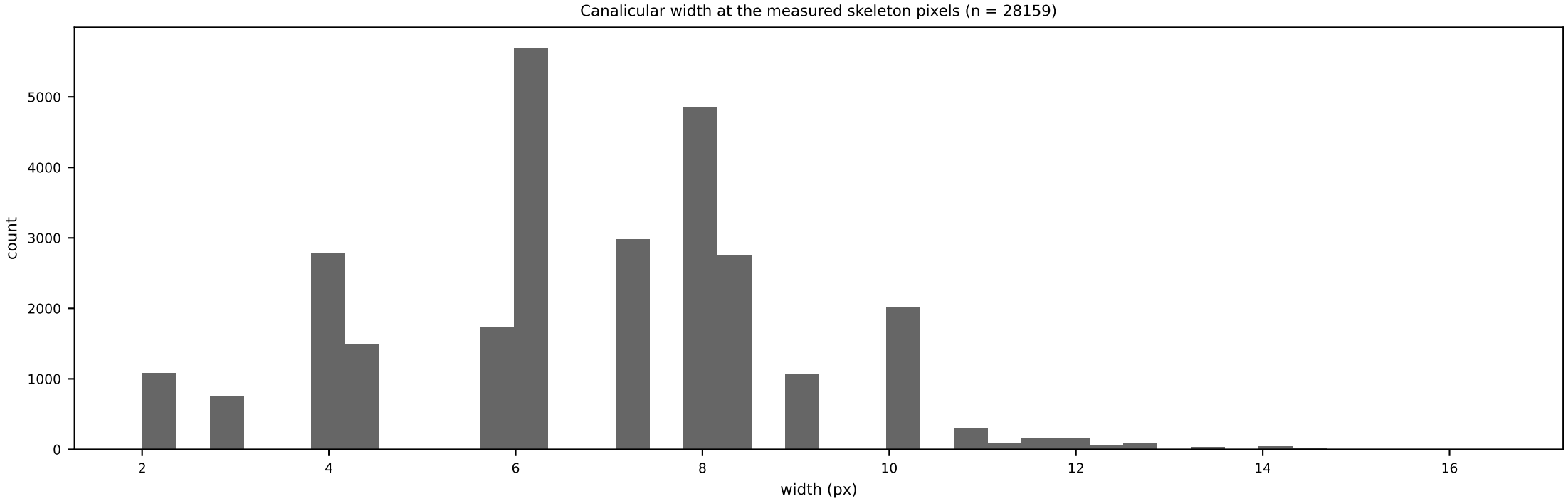
Normalised per lacuna

---

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	262.096	5607	13819	0.5809	0.5584	0.0228286	0.0166437	0.3815
2	229.522	9151	23871	1	0.999	0.0338761	0.0291567	3.4855
3	120.811	6134	17911	1	1	0.0428758	0.0316007	3.311
4	309.279	11331	28206	1	1	0.0321243	0.0418351	2.2633
5	235.238	9387	21888	1	1	0.0278044	0.0344024	1.7004
6	214.167	8637	20090	1	1	0.0439968	0.0410154	2.8016
7	322.836	9940	22198	0.8402	0.7724	0.0350101	0.033922	2.478
8	229.48	8595	22042	1	1	0.0305992	0.0325288	2.1788
9	109.498	4245	11066	1	0.9865	0.031331	0.0380445	4.5663
10	161.817	5491	13376	1	1	0.028228	0.0327452	2.4719
11	147.196	6914	19480	1	1	0.0170668	0.0197125	1.3587
12	254.693	9728	23664	1	1	0.0328947	0.0332995	2.7484
13	202.693	7711	16371	0.9376	0.8055	0.0243807	0.0214404	3.4535

Network v2 per lacuna

---

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	20	20	145.397	253.652	1	4	4
2	259	574	340.108	791.068	8	12	12
3	257	557	298.38	640.487	8	7	7
4	290	860	401.836	1313.55	10	13	16
5	160	334	279.61	824.186	5	12	12
6	311	599	424.978	927.625	7	13	14
7	337	649	389.664	844.34	10	12	12
8	126	186	289.551	790.642	7	11	10
9	119	296	148.125	468.019	6	4	5
10	133	298	180.095	489.676	4	7	8
11	46	80	137.439	446.073	2	5	8
12	220	423	350.593	873.997	6	13	15
13	135	211	206.681	390.019	6	5	6

682_z29: distributions, pre-validation, px